



Computer Addiction

TODO

2021-06-29

Computer Addiction Module

Module Summary

Ethics background required: Familiarity with the virtue ethics, deontological, and consequentialist ethical frameworks would be helpful. They are briefly reviewed for those who need the information

Subject matter referred to in this module:

Placement in overall ethics curriculum:

Academic year: Junior or Senior courses. Good candidate for course that covers software engineering.

Recommended previous modules: First year curriculum

Recommended follow-up modules: Nothing specific

Time required:

- Before class: 1 day technology fast, 10 minute reflection writeup
- In class: 30 minutes

Learning objectives:

- Have an experience of how their own compulsive device-use habits > affect their life.
- Understand the techniques used in making software products > addictive.
- Understand and critique the motivations for making a product > addictive.
- Explore the importance of supporting full human agency in the users > of their software. (Manipulation, human dignity)
- Identify alternative business models that don't rely on compulsively > habit-forming use.



Ethical issues to be considered: User behavior modification, human dignity

Exercise flow:

- Targeted technology fast
- Reflection on targeted technology fast
- View a video on building intentionally habit-forming software services.
- Discuss these techniques and the ethics of intentional psychological manipulation (both positive and negative).
- Discuss how can virtue ethics, deontological ethics, or consequentialism bring insight to the ethicality of this approach.
- Consider a product that was found to be addictive in the first activity: What alternatives can be imagined that would keep the product useful, but less addictive?

Preparation: Read the entire module and print/post copies of reliability scenarios to be distributed to groups. The technology fast activity should be assigned prior to the lesson.

Instructors Guide

Lesson plan

Technology Fast Homework

1 week before the discussion, hand out the following activity prompt

"Please choose one day this week where you will "give up" (entirely refrain from using) some tech device, app, or activity for just one day (eg. social media, gaming, texting etc.). The activity should be one that you use regularly in your daily life, but not one that would imperil your academic success in other courses. In your reading response, use your "personal reflection" section to comment on the experience of this small "technology fast". Was it difficult to do? How did it make you feel? What did you find your thoughts turning to? Would you consider making a "technology fast" part of your regular life rhythms? What is it about the design of this product that causes it to feel so compelling?"

- Student should submit their choice of particular technological > activity and a particular day before the fast.
- Student should complete the technology fast and submit their > reflection write-up at least one day prior to the class > discussion.

In-class Activity (30 min)



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(5 minutes) Invite student comments on the results of the technology fast.

(2 minutes) Watch the following video, promoting a book on how to make software habit-forming:

https://youtu.be/qWEldj_jj9E

(For a longer treatment of this topic, you may want to consider the documentary *The Social Dilemma*.)

(10 minutes) Discuss both the techniques described in the promotion, and the ethics of using those techniques. Draw out from the students the situations in which these techniques may be appropriate, and those in which they would be inappropriate.

Have the students identify which ethical frameworks they are using to make these judgments. (Virtue ethics, deontology, consequentialism)

(10 minutes) Broadening the discussion

Virtue Ethics:

What virtues are applicable when confronted with this issue?

How are habits and virtue linked?

Deontological Ethics:

Are there rules in the ACM code of ethics that would address this topic?

Consequentialism:

Who are the stakeholders? Who benefits? Who is negatively affected?

What are the societal impacts if this technology becomes ubiquitous?

If your boss asked you to design software using addictive techniques, how would you respond? (What other design choices or even business models could be suggested that don't have these concerns? How could a different design give the user of the product more control over their own behavior?)

Assessment (to be included at a later date on a quiz, paper, or exam to determine if LOs were reached)

- What ideas did the class come up with for responses to the boss's > request to develop an addictive product?
- What changes are you considering in your life after your technology > fast and/or hearing comments from your classmates?



- Using one or more of the Ethical Frameworks, describe why designing > products to be intentionally addictive is ethically problematic?

Student handouts are attached

U.S Patriot Missile Defense Scenario

In February of 1991, a U.S. Patriot missile defense system in Saudi Arabia, failed to detect an attack on an Army barracks. A government report found that a software problem led to an inaccurate tracking calculation that became worse the longer the system operated.

<https://raygun.com/blog/costly-software-errors-history/>

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Answer the following questions with your group

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 - Does the unreliability of the software you are considering cause > moral values (virtues) of you or your society to be violated? > Virtues include:
 - Honesty, integrity, compassion, courage, excellence, > dignity, justice, kindness, patience, quality, > perseverance, respect, service, transparency, etc.
 - Does the unreliability of the software you are considering cause > harm to people without producing great value to many more? (a > tradeoff)
 - Is there a rule that would be violated by allowing this software > to continue to be used knowing its faults?

Be prepared to discuss your answers with the whole group

U.K.’s Child Support Agency Scenario

In 2004, EDS introduced a highly complex IT system to the U.K.’s Child Support Agency (CSA). At the exact same time, the Department for Work and Pensions (DWP) decided to restructure the entire agency. The two pieces of software were completely incompatible, and irreversible errors were introduced as a result. The system somehow managed to overpay 1.9 million people, underpay another 700,000, had US\$7 billion in uncollected child



support payments, a backlog of 239,000 cases, 36,000 new cases “stuck” in the system, and has cost the UK taxpayers over US\$1 billion.

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Y2K Problem Scenario

The Year 2000 problem, also known as the Y2K problem, the Millennium bug, Y2K bug, the Y2K glitch, or Y2K, refers to events related to the formatting and storage of calendar data for dates beginning in the year 2000. Problems were anticipated, and arose, because many programs represented four-digit years with only the final two digits – making the year 2000 indistinguishable from 1900. The assumption of a twentieth-century date in such programs could cause various errors, such as the incorrect display of dates and the inaccurate ordering of automated dated records or real-time events.

https://en.wikipedia.org/wiki/Year_2000_problem

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NASA's Mars Climate Orbiter Scenario

On its mission to Mars in 1998 the Climate Orbiter spacecraft was ultimately lost in space. Although the failure bemused engineers for some time it was revealed that a sub contractor on the engineering team failed to make a simple conversion from English units to metric. An embarrassing lapse that sent the \$125 million craft fatally close to Mars' surface after attempting to stabilize its orbit too low. Flight controllers believe the spacecraft plowed into Mars' atmosphere where the associated stresses crippled its communications, leaving it hurtling on through space in an orbit around the sun.

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Ariane 5 Flight 501 Scenario

Europe's newest un-manned satellite-launching rocket reused working software from its predecessor, the Ariane 4. Unfortunately, the Ariane 5's faster engines exploited a bug that was not found in previous models. Thirty-six seconds into its maiden launch the rocket's engineers hit the self destruct button following multiple computer failures. In essence, the software had tried to cram a 64-bit number into a 16-bit space. The resulting overflow conditions crashed both the primary and backup computers (which were both running the exact same software). The Ariane 5 had cost nearly \$8 billion to develop, and was carrying a \$500 million satellite payload when it exploded.

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Heathrow Terminal 5 Opening Scenario

Just before the opening of Heathrow's Terminal 5 in the UK, staff tested the brand new baggage handling system built to carry the vast amounts of luggage checked in each day. Engineers tested the system thoroughly before opening the Terminal to the public with over 12,000 test pieces of luggage. It worked flawlessly on all test runs only to find on the Terminal's opening day the system simply could not cope. It is thought that "real life" scenarios such as removing a bag from the system manually when a passenger had left an important item in their luggage, had caused the entire system to become confused and shut down. Over the following 10 days some 42,000 bags failed to travel with their owners, and over 500 flights were cancelled.

<https://www.computerworld.com/article/3412197/top-software-failures-in-recent-history.html>

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Financial Software Scenario

A customer using a financial software package preparing for an upcoming client meeting. The financial advisor is reviewing his presentation and recognizes that the investment



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earning for his client are showing a negative value when he is certain that the investments have made a gain. The financial advisor calls the software company asking if he has used the software incorrectly or if there is a known error. The customer service representative says that she is not aware of a known error, but will ask the maintenance programmers to look into the problem. The programmers had a very difficult time finding the bug, but a week later they located it. Unfortunately, in the absence of the software, the financial advisor had to reschedule and lost the client due to the delay.

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Accounting System Scenario

A computer company is writing the first stage of a more efficient accounting system which will be used by the government and will save taxpayers a considerable amount of money. A software engineer is asked to design a user interface for the system. The accounting system and the interface contain all of the functionality described in the requirements. The system is installed, but the user interface is so hard to use that the complaints of the customer's staff are heard by the customer's upper level management. Because of these complaints, upper level management will not invest any more money in the development of the new accounting system and they go back to their original more expensive system.

<https://csciwww.etsu.edu/gotterbarn/artge2.htm>



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Radar System Scenario

A computer company wrote a radar system for ships which identifies incoming aircraft as friend or foe. A software engineer developed a user interface which will display needed information. This information can be used to determine if defensive action is necessary. The radar system and the interface contain all of the functionality described in the requirements. Although all of the information required was displayed, it was not displayed in a way that was adequate for them to distinguish a military attack plane from a commercial airliner. This contributed to the loss of many lives.

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Junior Programmer Scenario

One of the main clients of ACME Programming Associates is Always Airways. Always Airways has employed ACME to create simulators for training their pilots on various aircraft. One simulator needs to be updated because the 2020 version of commuter planes used by Always contains an air circulation monitor that was not previously included on the cockpit dashboard. The updated simulator needs to be ready for training pilots next week. They are on a very strict timeline but the only programmer available to ACME is Kim, a fairly new college graduate who has only been with the company for 2 months and just finished her training on the company's proprietary IDE. Since the addition of the monitor to the dashboard should be a fairly simple change, Kim is assigned to the job. Kim tested the addition to the best of her ability, but did not realize that the units used for air circulation could produce values up to 100 billion. Hence, when the simulator was used by the pilots, negative numbers appeared.

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